Data 275P: Fall 2026

Machine Learning with Generative Models

Homework 1

Due Date: **Friday, October 9th**

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Welcome to CS 179!

The homeworks in this course consist of a mixture of manual / handwritten problems, and code / data processing. To help simplify the grading process for the class, we will submit our homework solutions to gradescope via a **templated PDF**. In subsequent homeworks, there may be an additional submission process for providing code used in the assignment, as well.

Canvas includes detailed guidelines on how homework solutions should be formatted for submission to gradescope. In particular, please take care that:

- For the PDF submission, you must **fill in this PDF template** to provide your answers. All pages should remain in their original order when uploading your assignment to gradescope. *Do not rearrange, add, or remove any pages from the provided PDF template.*
- Enter your final answers in the dedicated answer spaces, and write your explanations and/or any math equations in the space provided below each question. *Always show the work needed to produce your answers, not just the final result.* Although typed responses are preferred, legible handwritten responses are fine.
- For multiple-choice questions, be sure to clearly select one of the answers, as well as providing an explanation or justification to support your answer.
- Check that all answers are readable, all scanned math or imported images are clear, and that content has not been cut off by the page margins. (We can only grade answers that we can see!)
- A Jupyter notebook version of the homework is also provided with each assignment.
- Some responses require inclusion of a plot or figure; after generating the plot in python, you can export it to a file or copy it to your clipboard, and then paste it into the homework PDF where indicated.
- If you would like to include code snippets in your responses, you can do the same thing – take an image of your code, and import it into the PDF in the space provided, or paste it into a textbox in the PDF.

If you have any questions/concerns about the homework problems or using Jupyter notebooks, ask us on EdD.

Summary of Assignment: 100 total points

- Problem 1: Dentist calculations (20 points)
- Problem 2: Hypothesis Testing (30 points)
- Problem 3: Naïve Bayes (25 points + 25 points [code])

Problem 1: Dentist Calculations (20 points)

Consider the joint probability distribution for the dentist example given in class (see also slides):

T	D	C	$p(T,D,C)$
0	0	0	0.576
0	0	1	0.008
0	1	0	0.144
0	1	1	0.072
1	0	0	0.064
1	0	1	0.012
1	1	0	0.016
1	1	1	0.108

$P(C) = 0.008 + 0.072 + 0.012 + 0.108 = 0.2$

$P(T=1, D=0) = 0.064 + 0.012 = 0.076$

Calculate the following marginal or conditional probabilities, showing your work:

- $P(D=1)$, the probability of the dental probe catching.

$P(D=1) = \sum_T \sum_C P(T,D,C|D=1)$

$= 0.144 + 0.072 + 0.016 + 0.108$

$= 0.34$

$P(D=1) =$ 0.34

- $P(C=1, D=1)$, the probability of having a cavity and the probe catching.

$0.072 + 0.108$

$P(C=1, D=1) =$ 0.18

- $P(D=1|C=1)$, the probability of the probe catching given that we have a cavity.

$= \frac{P(D=1, C=1)}{P(C=1)}$ $P(C=1) = 0.2$

$P(D=1, C=1) = 0.18$

$\frac{0.18}{0.2}$

$P(D=1|C=1) =$ 0.9

- $P(C=1|T=1, D=0)$, the probability that we have a cavity given that we have a toothache but the probe does not catch.

$= \frac{P(C=1, T=1, D=0)}{P(T=1, D=0)} = \frac{0.012}{0.076}$

$P(C=1|T=1, D=0) =$ ≈ 0.1579

- Are T and D independent? How do you know? Give a numerical justification.

  $T \perp\!\!\!\perp D$? Why?

If $T \perp\!\!\!\perp D$, $P(T|D) = P(T)$

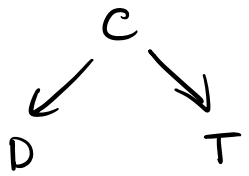
$P(T|D) = \frac{0.124}{0.34} \approx 0.3647$

$P(T) = 0.2$

$P(T|D) \neq P(T)$, Therefore, $T \not\perp\!\!\!\perp D$

- We know from class that D and T are *conditionally independent* given C. Using this information, sketch a directed graphical model that represents $p(T, D, C)$.

  Bayesian network:



Problem 2: Hypothesis Testing (30 points)

You've been asked to test the performance of a batch of newly fabricated processors. If the processors were correctly manufactured (class $T = 0$), the time X to complete your test suite is exponentially distributed with mean 1. If the equipment at the factory malfunctions (class $T = 1$), the time X is exponentially distributed with mean 50. Recall that the exponential probability density function equals $p(x) = \theta \exp(-\theta x)$ for $x \geq 0$, where $\mathbb{E}[X] = 1/\theta$.

You must decide whether or not this batch of processors was correctly manufactured. For the scenarios in the three parts below, it is possible to show that the optimal Bayesian classifier predicts $T = 0$ if $x \leq c$, and predicts $T = 1$ if $x > c$, for some constant c. The value of c depends on the test time distributions, the prior probabilities of the two classes, and the assumed loss function. You need to determine the optimal c in each case.

- Suppose that a new fabrication process has just been deployed, and the probability that the factory manufactures correctly functioning processors is only $P(T = 0) = 0.5$. What threshold c of the observed test suite time $X = x$ maximizes the probability that your prediction is correct?

$P(x) = P(T=0)P(x|T=0) = P(T=1)P(x|T=1)$

$\hookrightarrow 0.5e^{-c} = 0.5(0.02)e^{-0.02c}$

$e^{-c} = 0.02e^{-0.02c}$

$\ln(e^{-c}) = \ln(0.02e^{-0.02c})$

$-c = \ln(0.02) - 0.02c$

$-\ln(0.02) = 0.98c$

$3.912 = 0.98c$

$c = 3.99$

$c =$ 3.99

- Now suppose that after some improvements to the new fabrication process, the probability that the factory manufactures correctly functioning processors increases to $P(T = 0) = 0.99$. What threshold c of the observed test suite time $X = x$ maximizes the probability that your prediction is correct?

$P(T=0) = 0.99 \iff P(T=1) = 0.01$

$0.99e^{-c} = 0.01(0.02)e^{-0.02c}$

$\ln(0.99) - c = \ln(0.0002) - 0.02c$

$-0.01 - c = -8.52 - 0.02c$

$8.51 = 0.98c$

$c = 8.68$

$c =$ 8.68

- Market research suggests that the loss (or cost) of a missed detection (predicting $T = 0$ when the processor is actually defective) is 500 times greater than the loss of a false alarm (predicting $T = 1$ when the processor was correctly manufactured). Assuming again that $P(T = 0) = 0.99$, what threshold c of the observed test suite time $X = x$ minimizes the expected loss?

$L(\text{missed}) = 500\,L(\text{false alarm})$

$500 \cdot 0.01 \cdot 0.02e^{-0.02c} = 0.99e^{-c}$

$0.1e^{-0.02c} = 0.99e^{-c}$

$\ln(0.1) - 0.02c = \ln(0.99) - c$

$0.98c = \ln(9.9)$

$= 2.29$

$c = 2.34$

$c =$ 2.34

Problem 3: Naïve Bayes (25 points PDF; 25 points Python)

We consider data (produced by a sophisticated simulator) like that which would be collected by from a gamma telescope observing high energy particles. The raw data, "showers" of particles on a planar detector, have been converted into 10 continuous features as outlined here: `http://archive.ics.uci.edu/ml/datasets/MAGIC+Gamma+Telescope`. Our goal is the binary classification of the "primary" gamma signals of scientific interest from background, hadronic shower events. The $D = 10$ continuous features for each of the $N = 15,216$ training examples are stored in a $N \times D$ matrix `train`. The class labels are stored in an $N \times 1$ vector `trainLabels`, where primary gamma signals have label 1 and background events have label 0. Similarly, test data is stored in `test` and `testLabels`.

Let $x_d^{(n)} \in \mathbb{R}$ be the value of feature d for training example n, and $y^{(n)} \in \{0, 1\}$ the class label for example n. We will build a "naïve Bayes" classifier, which when all errors are equally costly predicts observation n to be a gamma signal when $p(y^{(n)} = 1|x^{(n)}) > p(y^{(n)} = 0|x^{(n)})$, and a background event otherwise. Using Bayes rule, this classifier is equivalent to one that predicts $y^{(n)} = 1$ if and only if:

$$\frac{p(y^{(n)} = 1)\, p(x^{(n)}|y^{(n)} = 1)}{p(x^{(n)})} \quad > \quad \frac{p(y^{(n)} = 0)\, p(x^{(n)}|y^{(n)} = 0)}{p(x^{(n)})},$$

or equivalently,

$$\log p(y^{(n)} = 1) + \log p(x^{(n)}|y^{(n)} = 1) \quad > \quad \log p(y^{(n)} = 0) + \log p(x^{(n)}|y^{(n)} = 0).$$

In this equation, $p(y^{(n)})$ is the prior probability of gamma signals and background events. The class conditional probability density function $p(x^{(n)}|y^{(n)})$ describes the distribution of the $D = 10$ features, which we assume depends on the type of event. We make two simplifying assumptions about these densities: the features $x_d^{(n)}$ are conditionally independent given $y^{(n)}$, and their distributions are Gaussian. Thus:

$$p(x^{(n)}|y^{(n)} = 1) = \prod_{d=1}^{D} \frac{1}{\sqrt{2\pi\sigma_{1,d}^2}} \exp\left[-\frac{(x_d^{(n)} - \mu_{1,d})^2}{2\sigma_{1,d}^2} \right], \tag{1}$$

$$p(x^{(n)}|y^{(n)} = 0) = \prod_{d=1}^{D} \frac{1}{\sqrt{2\pi\sigma_{0,d}^2}} \exp\left[-\frac{(x_d^{(n)} - \mu_{0,d})^2}{2\sigma_{0,d}^2} \right], \tag{2}$$

Given $y^{(n)} = 1$, each feature $x_d^{(n)}$ is Gaussian with mean $\mu_{1,d}$ and variance $\sigma_{1,d}^2$. Given $y^{(n)} = 0$, each feature $x_d^{(n)}$ is Gaussian with mean $\mu_{0,d}$ and variance $\sigma_{0,d}^2$. There are a total of $2D$ mean parameters and $2D$ variance parameters, since every feature has a distinct distribution for each of the two classes.

- Derive equations for $\log p(x^{(n)}|y^{(n)}=1)$ and $\log p(x^{(n)}|y^{(n)}=0)$, the (natural) logarithms of the conditional probability density functions in Equations (1)–(2). For numerical robustness, simplify your answer so that it does not involve the exponential function.

$\log p(x^{(n)}|y^{(n)}=1) =$

$$\sum_{d=1}^{n}\left[-\frac{1}{2}\log(2\pi\theta_{1,d}^2)-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\theta_{1,d}^2}\right]$$

$\log p(x^{(n)}|y^{(n)}=0) =$

$$\sum_{d=1}^{n}\left[-\frac{1}{2}\log(2\pi\theta_{0,d}^2)-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\theta_{0,d}^2}\right]$$

$$\log p(x^{(n)}|y^{(n)}=1) = \log\left(\prod_{d=1}^{D}\frac{1}{\sqrt{2\pi\sigma_{1,d}^2}}\exp\left[-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\sigma_{1,d}^2}\right],\right)$$

$$= \sum_{d=1}^{n}\log\left(\frac{1}{\sqrt{2\pi\theta_{1}^2}}e\left[-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\theta_{1,d}^2}\right]\right)$$

$$= \sum_{d=1}^{n}-\frac{1}{2}\log(2\pi\theta_{1}^2)+\log\left(e\left[-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\theta_{1,d}^2}\right]\right)$$

$$= \sum_{d=1}^{n}\left[-\frac{1}{2}\log(2\pi\theta_{1,d}^2)-\frac{(x_d^{(n)}-\mu_{1,d})^2}{2\theta_{1,d}^2}\right]$$

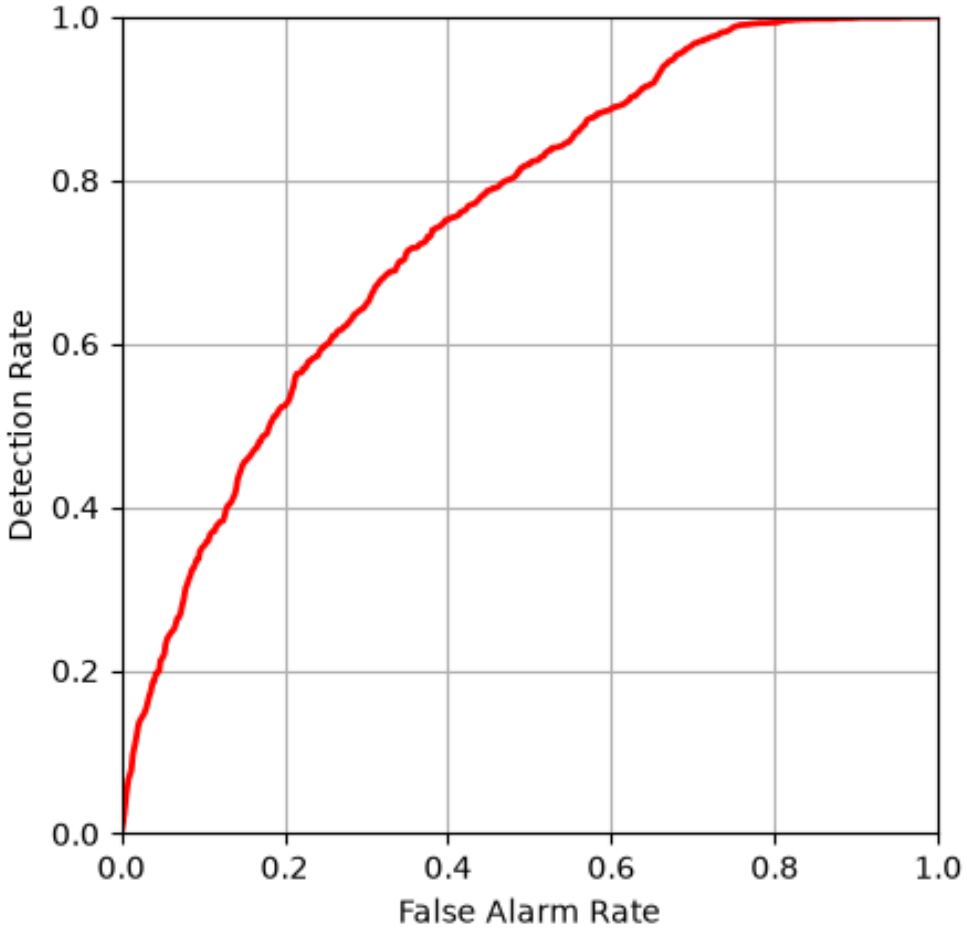
- **Implement code** that computes maximum likelihood (ML) estimates of the parameters of the Gaussian naïve Bayes model: the class-conditional feature means $\mu_{1,d}, \mu_{0,d}$; the class-conditional feature variances $\sigma_{1,d}, \sigma_{0,d}$; and the priors $q = p(y^{(n)}=0), 1-q = p(y^{(n)}=1)$. Hint: The demo code shows how to compute the ML estimates of the feature means. You do not need to report anything in your PDF answers for this part.

- Using your results from part (a), and the ML parameter estimates from part (b), write code that evaluates the log-likelihood ratio $\log p(x^{(n)}|y^{(n)}=1) - \log p(x^{(n)}|y^{(n)}=0)$ for every test data point. Using these "confidence" scores, compute and plot an ROC curve to evaluate your classifier's test performance. Also report the classifier accuracy on the test data.

Test Accuracy

0.7290

ROC Curve



- Suppose that the frequencies of the classes are as in the training data, and that all errors are equally costly. Classify each test example according to the optimal Bayesian classification rule. What are the true positive rate and false positive rate on the test data?

True Positive Rate

0.9104

False Positive Rate

0.6354

- Suppose that the frequencies of the classes are as in the training data, and that it is 50 times more costly to classify signals as background (missed detections) as to classify background as signals (false alarms). Classify each test example according to the optimal Bayesian classification rule. What are the true positive rate and false positive rate on the test data?

True Positive Rate

0.9673

False Positive Rate

0.7009